

AUTO CFD WORKSHOP SERIES

AutoCFD5 AIML DrivAerML

Participant Submission Instructions

A reproducible route to export complete native-cell surface predictions, optionally include complete volume predictions, run the approved evaluator, inspect results, and deliver a verified entry.

PUBLIC TOOLING

Evaluator code, documentation, immutable profile support, and the report-only regional contract.

CONFIDENTIAL ENTRIES

Participant result packages use a private upload-only route.

Version 1.1.4 / 31 August 2026

<https://github.com/neilashton/autocfd5-aiml-submission>

The complete route, at a glance

You run the official evaluator on your own machine using the pinned DrivAerML dataset. The public repository carries the code and fixed support data; your predictions and results remain under your control until you use the private workshop upload route. The next section lists every official split.

Mandatory in both scopes: the complete native surface

For **every selected test case**, export `pMeanTrim` and `wallShearStressMeanTrim` for **every cell** of the pinned surface VTP. Inference may run in chunks or on another representation, but the final surface export must map back to every native `raw_cell_id` exactly once, with no missing or duplicate cells. A sampled or partial surface is not accepted.

prediction_scope	Required predictions	Maximum overall score
surface_and_volume	Complete native surface plus complete native-volume <code>pMeanTrim</code> and <code>UMeanTrim</code> . This is the legacy default.	100 / 100
surface_only	Complete native surface; omit volume predictions. Volume fields and velocity profiles receive zero component points.	60 / 100

Surface-only entries do not use dummy volume fields. Their unavailable scientific metric values are absent, the three unavailable component scores are exactly zero, and the approved weights are not renormalized.

- 01 Clone the evaluator and select the frozen evaluator-v1.1.4 release.
- 02 Fetch the immutable profile-support bundle and the pinned native test files.
- 03 Choose one prediction scope in `entry.json` and export every required native cell for every selected case.
- 04 Validate the entry, then evaluate one case while developing your export.
- 05 Evaluate the official full split as the minimum common comparison.
- 06 Inspect `result.json` and selected local HTML profile reports.
- 07 Create and verify one deterministic ZIP, then upload it confidentially.

Public repository does not mean public submissions

Do not commit predictions or results, attach them to an issue, or open a pull request. The organisers provide an AutoCFD Dropbox File Request for confidential upload. The email is only a checksum receipt.

Who is responsible for what

Party	Responsibility
Participant	Export native-order predictions; run, inspect, package, verify, upload, and retain the original ZIP and checksum.
Evaluator	Verify fixed inputs; calculate fields, forces, and profiles; aggregate the full split; write compact, deterministic outputs.
Organisers	Provide the private upload link; verify package identity and split completeness; preserve the received ZIP under the workshop embargo.

What you hand in

- One verified `.zip` produced by `autocfd5-aiml` package.
- A short email containing the committee-issued submission ID, filename, and exact SHA-256.
- No raw native prediction files unless the organisers explicitly request a separately hosted immutable artifact.

Use Full as the common baseline

The AutoCFD organising committee asks every participant to evaluate the official Full split as the minimum common comparison. You are welcome to test the other official splits too.

Committee recommendation

Use `split_id = full` for the primary submission. Official split membership is frozen in the evaluator, so participants do not need to repeat its training and validation IDs in `entry.json`.

Official split	Train	Validation	Test	Case set
full	400	34	50	Standard - requested baseline
medium	133	34	50	Standard
scarce	67	34	50	Standard
super_scarce	11	34	50	Standard
geometry	339	48	97	Geometry
high_drag	339	48	97	High drag
low_drag	339	48	97	Low drag
rear_separation	339	48	97	Rear separation

If you use a custom split

The organisers strongly recommend the official splits. If an additional split is not one of those above, give it a new safe `split_id` and include all three ordered run-ID arrays in `entry.json`:

```
{
  "split_id": "my-custom-study",
  "train_case_ids": ["run_1", "run_2"],
  "validation_case_ids": ["run_3"],
  "test_case_ids": ["run_4", "run_5"]
}
```

- Each array must be non-empty, unique, use `run_N` IDs from the pinned dataset, and be disjoint from the other two arrays.
- The evaluator writes `custom-split.json` into the verified result package for the organisers.

```
autocfd5-aiml fetch-data --entry-root my-entry \
  --destination /data/drivaerml --dry-run
```

03 / ENVIRONMENT

Install once; pin everything

Run the scientific calculation on Linux with Python 3.12. The evaluator pins NumPy and VTK because native-file handling and numerical reproducibility are part of the contract.

Required environment

Linux; Python $\geq 3.12, < 3.13$; NumPy 2.2.6; VTK 9.5.2; Git; GitHub CLI (gh); and Hugging Face CLI (hf). A Dockerfile using Python 3.12.13 is included when a clean container is preferable.

Clone and install

```
git clone https://github.com/neilashton/autocfd5-aiml-submission.git
cd autocfd5-aiml-submission
git checkout evaluator-v1.1.4

python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -e '.[test]'
autocfd5-aiml --help
```

Fetch and verify profile support

```
autocfd5-aiml fetch-support --destination support/native-v1
```

The command downloads the **support-v1** release asset, verifies its archive hash, extracts it safely, and verifies **index.json**. It refuses a non-empty destination.

Inspect the native-data download before starting it

```
autocfd5-aiml fetch-data \
  --split-id full \
  --prediction-scope surface_and_volume \
  --destination /data/drivaerml \
  --dry-run

# Remove --dry-run only when the location and required storage are ready.
autocfd5-aiml fetch-data \
  --split-id full \
  --prediction-scope surface_and_volume \
  --destination /data/drivaerml
```

The native files are very large. The dry run lists the exact pinned files and sizes. Replace `surface_and_volume` with `surface_only` to omit native volume parts, or use `--entry-root my-entry` after creating the entry so the command reads its exact cases and scope. The final dataset root must contain `force_mom_constref_all.csv` and each selected `run_N/` directory in the downloaded layout.

Immutable identities

Dataset	neashton/drivaerml
Dataset revision	7a5c0948ce27be709b1116a3a190f806e7a8f79f
Support release	support-v1
Support ZIP SHA-256	5ebcf744be53016bd158236d1f4af3290ff399b323c0e11a49c37ea9a6c686f6
Support index SHA-256	f47f8c3ed7a56632b0c02a3aec793e4cd823d5d04d5264d00fcd419bf11c0f4f
Regional report contract SHA-256	2bfd372817989112642056e4c76cfb418dbdcee445c57ee20ca37ee9ca158583

Declare the entry and exact split

Use the exact submission ID sent to you by the AutoCFD organising committee. Start from the supplied Full-split example and keep its test-case membership and order unchanged.

Create a working entry

```
cp -R examples/entry my-entry
```

entry.json key	Rule
schema	Keep autocfd5-aiml-entry-v1.
schema_version	Keep integer 1.
submission_id	Enter the exact ID sent by the AutoCFD organising committee. It uses lowercase letters, digits, dot, dash, or underscore; at most 80 characters.
method_name	Human-readable method name, 1-200 characters.
contact_email	Email monitored by the participant.
split_id	Keep full for the requested baseline.
prediction_scope	Choose surface_and_volume or surface_only. State it explicitly; omission retains legacy full-field behaviour.
test_case_ids	Copy the example exactly. Membership and order are both checked.
prediction_artifact	Optional; only for an organiser-requested private immutable raw artifact.

Minimal shape

```
{
  "schema": "autocfd5-aiml-entry-v1",
  "schema_version": 1,
  "submission_id": "assigned-submission-id",
  "method_name": "Method display name",
  "contact_email": "participant@example.org",
  "split_id": "full",
  "prediction_scope": "surface_and_volume",
  "test_case_ids": [ ...copy the complete example array exactly... ]
}
```

The ellipsis above is explanatory and is not valid JSON. Copy examples/entry/entry.json, which contains all 50 IDs.

Validate before producing every case

autocfd5-aiml validate-entry my-entry

Validation has two layers

validate-entry checks metadata and split declarations. For an official split it requires the frozen test membership and order; for a custom split it requires complete, disjoint, known train, validation, and test IDs. The full evaluation requires a valid complete surface manifest for every test case and, only for surface_and_volume, a valid complete volume manifest.

05 / PREDICTIONS

Export native cells, without reordering

Every case contains a complete surface manifest plus one or more compressed NPZ chunks. Full-field entries also contain a complete volume manifest. Raw cell IDs restore native order and must cover every supplied native-cell range exactly.

```
my-entry/  
  entry.json  
  cases/  
    run_11/  
      surface/  
        manifest.json  
        chunks/chunk-000000.npz  
      volume/          # surface_and_volume only  
        manifest.json  
        chunks/chunk-000000.npz  
    ...one directory for every case in the selected split...
```

Surface-only directory rule

With `prediction_scope = surface_only`, stop after the surface chunks and omit `volume/`. Do not create zero or dummy volume arrays. The evaluator records volume pressure, volume velocity, and velocity profiles as unavailable and gives those components zero points.

Required NPZ arrays

Association	Array	Dtype	Shape / meaning
Surface	raw_cell_id	int64	(rows,); native polygon ID
Surface	pMeanTrim	float32/64	(rows,); mean pressure
Surface	wallShearStressMeanTrim	float32/64	(rows, 3); wall-shear vector
Volume (full only)	raw_cell_id	int64	(rows,); native volume-cell ID
Volume (full only)	pMeanTrim	float32/64	(rows,); mean pressure
Volume (full only)	UMeanTrim	float32/64	(rows, 3); mean velocity vector

Closed-manifest requirements

- Surface association is `CellData` with support ID `surface_native_cells`.
- The manifest declares case ID, total row count, field component counts, and every chunk.
- Each chunk declares file, SHA-256, row count, and half-open raw-ID range.
- Values must be finite; unknown or missing keys fail validation.
- IDs must cover `[0, total_row_count)` once, with no gaps or duplicates.
- The complete surface is mandatory in both scopes; a sampled or partial surface fails validation.
- Use at most 1,000,000 rows per chunk unless the organisers publish another limit.

Copy the closed manifest structure from the repository's Native prediction format document. Preserve native resolution rather than interpolating to a convenient grid; the evaluator hashes each NPZ before use and fails if an input changes.

Prove one case before the full run

Use a selected split case such as run_11 to catch layout, dtype, hash, ID-coverage, and native-data problems before committing resources to all 50 cases.

Evaluate one case

```
mkdir -p output/dev

autocfd5-aiml evaluate-case \
  --case-id run_11 \
  --dataset-root /data/drivaerml \
  --support-root support/native-v1 \
  --surface-manifest my-entry/cases/run_11/surface/manifest.json \
  --volume-manifest my-entry/cases/run_11/volume/manifest.json \
  --output output/dev/run_11.json
```

For a surface-only development check

```
autocfd5-aiml evaluate-case \
  --case-id run_11 \
  --prediction-scope surface_only \
  --dataset-root /data/drivaerml \
  --support-root support/native-v1 \
  --surface-manifest my-entry/cases/run_11/surface/manifest.json \
  --output output/dev/run_11-surface-only.json
```

A one-case result is diagnostic, not an official result

It contains every metric available in the declared scope, integrated force coefficients, and available profile loss statistics for that case. Force and profile R2 require variation across the complete test split, so a single case cannot produce official component or overall scores.

Check the result deliberately

Check	Expected outcome
Native identities	Every required boundary, area, and full-field volume part matches its pinned size and SHA-256.
Prediction coverage	Surface IDs are complete and unique; full-field entries also have complete unique volume IDs.
Numerics	All prediction values and all written JSON numbers are finite.
Forces	Cd, Cl, and pitch moment are integrated from the predicted native surface fields.
Profiles	Forty identities are retained. Surface-only entries mark all 32 velocity rows unavailable without prediction values; Cp remains available.
Regional reports	Surface regions are always present; volume regions appear only for full-field entries. All have zero scoring weight.
Gaps	Unsupported intervals remain separate segments; no line is drawn or integrated across them.

When the output already exists

The evaluator refuses to overwrite a case result. This is intentional. Use a new output filename after changing predictions, so every test result remains attributable to one input set.

07 / FULL EVALUATION

Run the complete split and inspect it

Official R2 values and the composite score are calculated only after every case in the selected test split has completed. The requested Full baseline contains 50 test cases. The declared prediction scope applies consistently to every case in that entry.

Run all cases

```
autocfd5-aiml evaluate-entry my-entry \
  --dataset-root /data/drivaerml \
  --support-root support/native-v1 \
  --output output/assigned-submission-id \
  --resume
```

If the process is interrupted, repeat the same command with `--resume`. Completed per-case work is reused only after its schema, case ID, and completion state pass validation. After `result.json` exists, that output is complete and immutable; use a new output directory for another model version.

Existing full-field inference is reusable

If you already produced complete v1.1.2 or v1.1.3 surface and volume prediction chunks, do not run model inference again. Keep those manifests and NPZ files, set `prediction_scope = surface_and_volume`, and rerun the v1.1.4 evaluator into a fresh output directory.

Output layout

```
output/assigned-submission-id/
  result.json           # aggregate metrics and identities
  regional-diagnostics.json # zero-weight regional field reports
  provenance.json       # runtime and verification record
  cases/run_N.json      # compact result for each test case
  profiles/index.json    # profile prediction index
  profiles/chunk-NNN.json # 40 series per case, eight cases per chunk
  .work/cases/run_N.json # resumable working data; excluded from package
```

Render a local case report

```
autocfd5-aiml report \
  --result-root output/assigned-submission-id \
  --support-root support/native-v1 \
  --case-id run_11 \
  --output output/run_11.html

# Open output/run_11.html in a browser on the same machine.
```

How to read the profile report

Series	Count/case	Role	Display and scoring behaviour
Constant velocity	16	Scored if volume	Raw U/U_inf samples for full entries; explicitly unavailable for surface-only.
Relative velocity	16	Report only	Full entries use the same numerical treatment; surface-only contains no prediction values.
Constant continuous Cp	4	Scored	Shown against physical x; integrated against arc length.
Relative Cp	4	Report only	Two aliases and two moving placements; zero composite weight.

Nine components, one approved composite

The composite keeps the approved nine fixed weights. Nominally, fields contribute 50%, integrated forces 25%, and constant-placement profiles 25%. Relative-placement profiles and regional field reports remain visible zero-weight diagnostics.

Group	Component metric	Overall weight	Transform / cap
Fields	Surface pressure relative L2, area weighted	15%	Error score; cap 15%
Fields	Surface wall shear relative L2, area weighted	10%	Error score; cap 20%
Fields	Volume velocity relative L2, equal native cells	15%	Error score; cap 12%
Fields	Volume pressure relative L2, equal native cells	10%	Error score; cap 15%
Forces	Cd R2	15%	Quality score
Forces	Cl R2	5%	Quality score
Forces	Pitch-moment R2	5%	Quality score
Profiles	16 constant velocity profiles: global weighted R2	15%	Quality score
Profiles	4 constant continuous Cp cuts: global weighted R2	10%	Quality score

Surface-only scoring: fixed 60-point ceiling

Surface pressure, wall shear, all three surface-integrated force components, and Cp remain available: together their fixed weights total 60%. Volume velocity (15%), volume pressure (10%), and constant velocity profiles (15%) are unavailable and each receives a transformed component score of exactly zero. Their scientific metric values are absent, no dummy fields are used, and no weight is transferred to another component. A perfect surface-only entry therefore scores 60/100.

Transforms

Field-error component

For relative L2 error e and declared cap c : $\text{score} = \text{clip}(100 \times (1 - e/c), 0, 100)$.

R2 component

For complete-split coefficient of determination R2: $\text{score} = 100 \times \text{clip}(R2, 0, 1)$. The weighted component scores sum to the overall score.

Important scientific details

- Available field errors are complete-case relative L2 percentages, macro-averaged equally across cases.
- Forces are integrated from predicted native surface pressure and wall shear. Pitch truth is $(Cl_f - Cl_r) / 2$.
- Velocity is $\text{magnitude}(U_{\text{MeanTrim}}) / 38.889$. Cp is $2 * p_{\text{MeanTrim}} / 38.889^2$.
- Profile integration follows each scoring coordinate. Every case/profile block is normalized to unit supported length, giving cases and profiles equal weight in global R2.
- Integration stops at every segment boundary. No unsupported interval is bridged and no smoothing is applied.
- Four-region surface reports, plus volume reports when submitted, reuse native-cell fields. They have weight 0.0 and are not consumed by any transformed component or the overall score.

09 / PACKAGE AND DELIVERY

Verify first; upload privately

The packaged result is compact and deterministic. It contains the declared scope, available scientific outputs, component availability and scores, available profile predictions, report-only regional diagnostics, entry identity, immutable input hashes, and runtime provenance - not the large raw native prediction chunks by default.

Create the result package

```
autocfd5-aiml package output/assigned-submission-id \  
  --output assigned-submission-id.zip  
  
autocfd5-aiml verify-package assigned-submission-id.zip
```

Replace assigned-submission-id with the exact ID sent by the AutoCFD organising committee. Packaging also writes assigned-submission-id.zip.sha256. Both commands refuse unsafe or inconsistent content. The ZIP is closed by package-manifest.json, which records every member's size and SHA-256.

Private hand-in

01 Open the AutoCFD Dropbox File Request.

02 Upload only assigned-submission-id.zip; wait for the upload to complete.

03 Retain the original ZIP and its generated .sha256 file unchanged.

04 Email the organisers the receipt details below. Do not attach the ZIP to email.

05 Keep predictions, results, and reports out of public Git history and issue trackers.

What the upload-only request protects

Participants can add their own file but cannot browse, download, replace, or compare other participants' submissions. Organisers keep the destination restricted to the processing group until the workshop reveal.

Checksum receipt email

```
Subject: [AutoCFD5 AIML] entry receipt - <submission-id>  
  
Submission ID: <submission-id>  
Method: <method-name>  
Uploaded filename: <submission-id>.zip  
SHA-256: <copy the 64-character value from <submission-id>.zip.sha256>  
Upload completed: <YYYY-MM-DD HH:MM UTC>  
Contact: <contact-email>
```

Optional large prediction artifact

Only when requested, store large native predictions at a private immutable URL and add prediction_artifact to entry.json with exact private_immutable_url, size_bytes, and sha256. The evaluator records the reference but does not copy that artifact into the result ZIP.

Questions: neil@neilashton.co.uk or astridwalle@cfdsolutions.net, the AutoCFD5 AI/ML TFG organisers.

Troubleshooting and final checks

Most failures are deliberate fail-closed checks. Correct the input or choose a fresh output location; do not edit a completed result package by hand.

Message or symptom	What it means / what to do
required command is unavailable	Install the named gh or hf CLI and confirm it is on PATH.
support destination is not empty	Use a new empty directory. Support extraction is intentionally non-overwriting.
archive, index, or native source differs	The file is not the pinned object. Re-fetch into a clean location and do not rename, recompress, or edit it.
entry split ID, order or membership differs	For an official split, restore its complete ordered test list. For a custom split, supply complete, disjoint train, validation, and test arrays.
prediction manifest or chunk identity differs	Regenerate manifest sizes and SHA-256 values after writing final NPZ chunks. Do not modify chunks afterward.
raw IDs are missing, repeated, or out of range	Export every native cell exactly once. Do not use solver-local reorderings without mapping back to raw native IDs.
volume manifest is required	The entry is using surface_and_volume (also the legacy default). Supply complete volume predictions or explicitly set prediction_scope to surface_only.
surface-only entry contains no velocity metric	Expected: velocity-profile and both volume scientific metrics are absent; their component scores are zero and weights stay fixed.
result.json already exists	That output directory is complete. Choose a new output directory for another model version.
interrupted full evaluation	Repeat the identical command with --resume; validated completed case work is retained.
profile line appears stepped or has a gap	This can be scientifically correct: raw samples are unsmoothed, and unsupported intervals are intentionally disconnected.
temporary space is exhausted	For full-field entries, the volume-region pass needs roughly 9 GiB of temporary local space per active case. Point TMPDIR at suitable local scratch or reduce evaluation concurrency. Surface-only never opens volume data.
regional diagnostics do not reconstruct	Do not edit the report. Re-run with the frozen evaluator and unchanged prediction chunks; all four exhaustive regions must reproduce the global additive field sums.

Final participant checklist

- [] I used Linux, Python 3.12, and the frozen **evaluator-v1.1.4** release.
- [] The support and native dataset identities verified automatically.
- [] My entry.json uses the submission ID sent by the AutoCFD organising committee and the official Full split.
- [] I declared one prediction scope. Every selected case has the complete native surface; full-field entries also have the complete native volume.
- [] evaluate-entry completed and result.json reports the exact split as complete.
- [] I inspected aggregate metrics and at least one local HTML report; any unavailable surface-only components are explicitly marked and contain no dummy values.
- [] I confirmed regional diagnostics are report only, have weight 0.0, and reconstruct every submitted field's unchanged global sums.
- [] verify-package reported the final ZIP as valid.
- [] I uploaded the ZIP through the AutoCFD Dropbox File Request and retained the original ZIP plus checksum.
- [] My receipt email contains the exact submission ID, filename, and SHA-256.